

quantum quadruple
Four computers for the price of one with a quantum computer, each quantum binary digit (qubit) is equivalent to four conventional on/off switches (bits)...



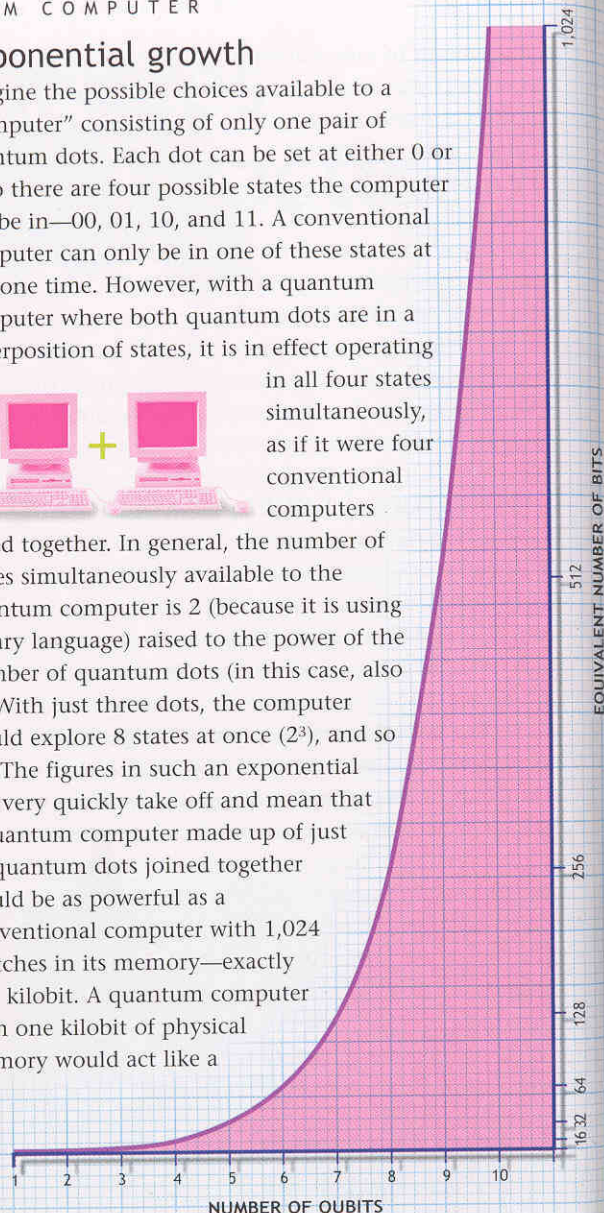
geometric increase
... but if you have twice as many bits, you don't double the power of the quantum computer—you square it. The power grows exponentially, so just 10 quantum dots (qubits) are equivalent to a current kilobyte.

exponential growth

Imagine the possible choices available to a "computer" consisting of only one pair of quantum dots. Each dot can be set at either 0 or 1. So there are four possible states the computer can be in—00, 01, 10, and 11. A conventional computer can only be in one of these states at any one time. However, with a quantum computer where both quantum dots are in a superposition of states, it is in effect operating

in all four states simultaneously, as if it were four conventional computers

wired together. In general, the number of states simultaneously available to the quantum computer is 2 (because it is using binary language) raised to the power of the number of quantum dots (in this case, also 2). With just three dots, the computer would explore 8 states at once (2^3), and so on. The figures in such an exponential law very quickly take off and mean that a quantum computer made up of just 10 quantum dots joined together would be as powerful as a conventional computer with 1,024 switches in its memory—exactly one kilobit. A quantum computer with one kilobit of physical memory would act like a



virtual machine with 10^{1024} bits of memory, a number vastly greater than the number of atoms in the Universe. It would be more powerful than a conventional computer made up of every atom in the visible Universe.

code-breaking

There are problems in learning how to connect the quantum dots in the right way, how to program such machines, and how to interpret their output. But solutions are being found with quantum computers made up of only a few dots, and physicists know that the effort is worthwhile because it has been proved mathematically

that full-sized quantum computers will work. A key feature of this highly technical proof came from the work of Peter Shor at the AT&T Bell Laboratories in New York.

Shor was interested in how quantum computers might be used to crack codes. There is a particular kind of code that is based on multiplying very large prime numbers together. The result is a much larger number, and the

“The fact that such extremely simple experiments can lead to such tremendous conclusions about the nature of reality, and the depth of those conclusions, is really the strangest thing I know. It’s strange, and it’s weird. But not, as many people say, mysterious—because I think we understand it.”

David Deutsch, Oxford physicist, (2000)



When **Peter Shor** (born 1959) looked into the problem of cracking codes that involve very

large prime numbers, he was able to show that a quantum computer could solve in seconds a problem that would take a conventional computer months. Now, all he has to do is build a quantum computer!

code can only be cracked by finding the two prime numbers that produced it—a process that is called factorizing. This process takes months, or even years using conventional computers. In 1994, Shor proved that a quantum computer could solve